Surge-Robust Flyback Power Supplies with GaN

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***Abstract*— In this paper, we have evaluated the surge robustness of a GaN FET-based Flyback power supply and demonstrated the maturity of GaN technology for adoption into high-volume markets. While the present generation of lateral GaN FETs do not have the capability to avalanche, significant transient overvoltage capability allows these devices to switch through surge events. Surge testing was based on the IEC 61000-4-5 and VDE 0884-11 standards, and we have shown in this work that GaN FETs survive the surge events without any hard failure or change in converter efficiency over 90V_{AC} - 264V_{AC} line conditions. GaN FETs have been shown to switch through with peak voltage over 800V, and for tens of milliseconds while exceeding the maximum continuously operating voltage rating of 650V. We have also demonstrated that GaN FETs can operate under momentaneous line voltage swell conditions per IEEE Standard 1159-2009 without suffering any degradation in converter efficiency. Such robustness enables the wide-spread adoption of GaN in cost-sensitive consumer electronics applications.**

Keywords*— *GaN power supply, Surge performance, GaN Flyback

I. Introduction

AC-DC adapters for consumer electronics including smart phones, tablets, laptops, and game consoles constitute one of the biggest markets for power supplies. Flyback conversion is the dominant topology for such low-power offline applications due to its simplicity and low cost. GaN-based power semiconductor devices are ideally suited for high-efficiency power converters due to their superior material properties [1]. GaN FETs are now being adopted into many commercial power supply designs [2-3], and devices are available from various manufacturers with voltage ratings up to 900V.

In order to have sufficient field reliability, it is essential that power supplies should withstand line-surge and transient over-voltage conditions observed during the course of typical real-life operation. Due to cost considerations, many commercial charger designs based on flyback topology do not include expensive protection features such as MOV's which can attenuate high voltages before reaching the power device. While the present generation lateral GaN FETs do not have the capability to avalanche, they do have significant transient overvoltage capability allowing them to switch through such events, and a methodology to validate this has been presented recently [4].

II. Flyback Power Supply with GaN

The schematic of a flyback power supply designed with a GaN FET that was evaluated in this work is shown in Fig 1.

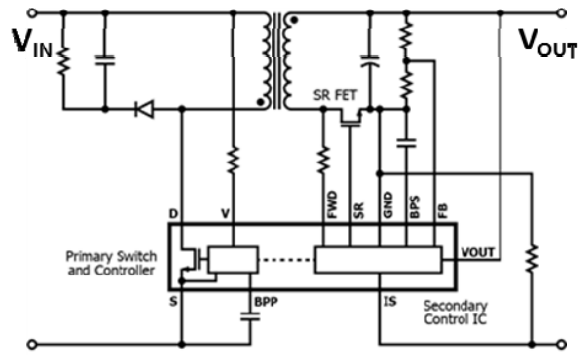

Fig. 1: Schematic of a QR flyback power supply with GaN FET

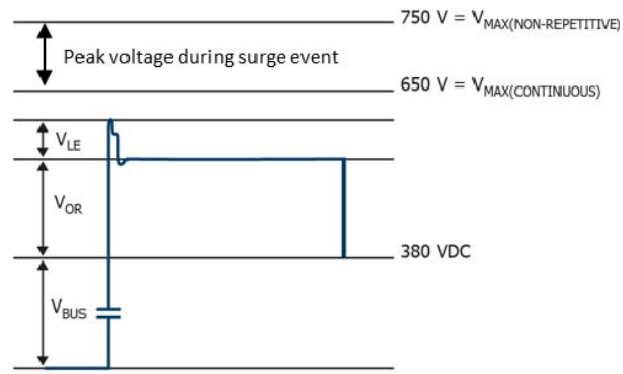

Fig. 2: Voltage stress on the primary switch in a flyback power supply at 264V_{AC}.

The tested part is an integrated flyback switcher IC which includes the GaN FET as the primary switch, primary and secondary side controllers and SR FET driver in a single surface mounted package, belonging to Innoswitch3 family of devices [5]. The GaN FET used in this product was fabricated in the internally developed PowiGaN^{TM} technology.

Voltage stress on the primary switch in a flyback power supply is illustrated in Fig. 2. It is a combination of the rectified line voltage (V_{BUS}), the output reflected voltage (V_{OR}) –the output voltage reflected across the turns-ratio of the power transformer, and the voltage induced by the leakage inductance of the transformer primary winding (V_{LE}). The device allows for the power supply to be designed with up to 650V across the GaN FET under normal operating conditions. When a power line surge occurs, the GaN FET has the capability to withstand a higher voltage, and this is indicated by the 750V maximum specified under non-repetitive conditions.

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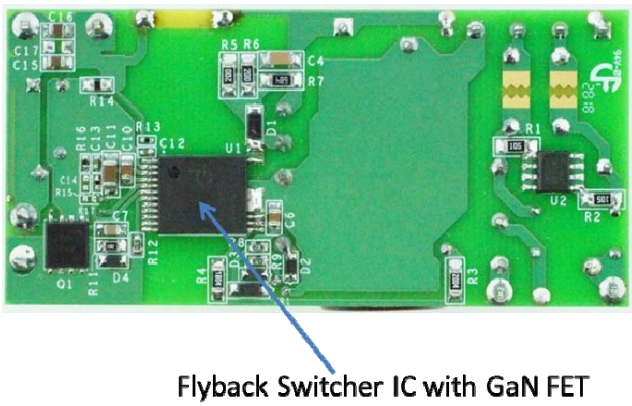

Fig. 3: Photograph of a populated circuit board used in this evaluation [6].

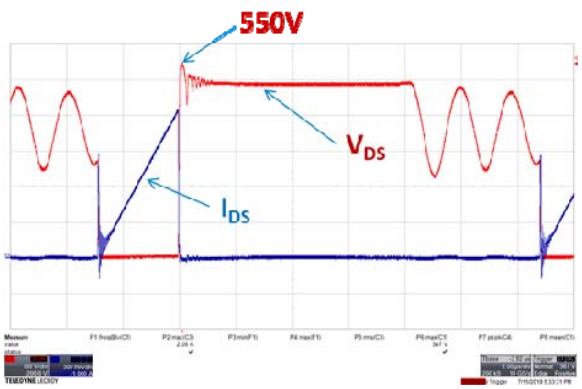

Fig. 4: Voltage and current waveforms on the GaN FET during typical operation at $264V_{AC}$ input.

It is essential that the GaN FET can switch through such a line surge event without suffering any permanent damage due to violation of its Safe Operating Area (SOA), or result in a degradation in subsequent performance.

III. Surge Performance Evaluation

The test vehicle used in this study is a 65W (20V/3.25A) power supply using Innoswitch3 GaN-based device as shown in Fig. 3, and its construction details are described in [6]. Voltage and current waveforms under typical operation at $264V_{AC}$ input is shown in Fig. 4. Under steady-state operating conditions, the peak drain voltage on the GaN FET is almost 550V.

The methodology for surge testing was based on the IEC 61000-4-5 standard [7] which specifies the surge generator output into both open and short circuits. The Combination Wave Generator 1.2/50µs open-circuit waveform shown in Fig. 5 was used in this evaluation and the schematic of the test set-up is illustrated in Fig. 6. The number of strikes was set at 50 per VDE 0884-11 standard [8]. A total of ten parts from one production lot were subjected to this fifty-event surge test.

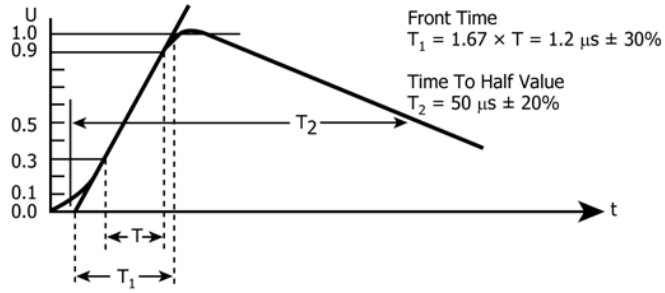

Fig. 5: IEC 61000-4-5 standard Combination Wave Generator (CWG) 1.2/50µs open-circuit surge waveform shape.

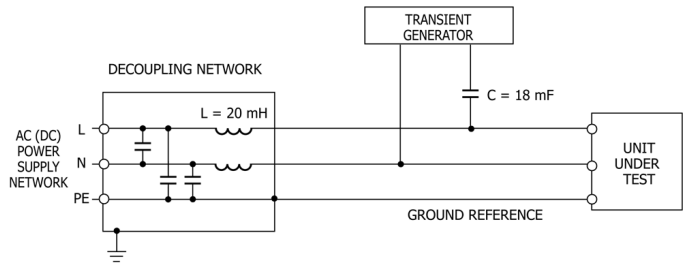

Fig. 6: Simplified test set-up used in this surge test evaluation.

The converter was operating with an input bus voltage of 375V ($264V_{AC}$), with a steady state peak drain voltage of 550V on the GaN FET when the surge strike was applied. A typical waveform captured during one such surge strike is shown in detail in Fig. 7, where the bottom half of the figure shows the zoomed image of the moment the strike is applied. Input bus voltage instantly rose with its peak reaching above 660V. As a result, the drain voltage across the GaN FET rose to 830V peak and 774V plateau immediately after the surge strike, going well above the 650V maximum continuous operation rating of the GaN FET. As seen in the top half of the figure, the GaN FET continued to switch through the surge event for tens of milliseconds with a peak drain voltage above 650V till the input bulk capacitor eventually discharged to its steady state level of 375V. This was repeated 50 times based on the VDE 0884-11 standard with no damage to any part that was stressed. This clearly demonstrates the capability of GaN FETs to switch through high voltage transient events without suffering any hard failure due to violation of the Safe Operating Area (SOA).

Another important parameter of interest is the converter efficiency – to understand if any performance degradation is observed after the surge strikes, caused by an increase in dynamic $R_{DS(ON)}$ of the GaN FET due to additional electron trapping during the high-voltage switching transitions accompanying the surge event. Any significant increase in the dynamic $R_{DS(ON)}$ would have a detrimental effect on the converter efficiency. Ten parts were subjected to the fifty-event surge test, and the converter efficiency was monitored both before and after the application of 50 surge strikes. Efficiency was normalized prior to stress (at 0 strikes) for every part tested, and the change in efficiency after the application of 50

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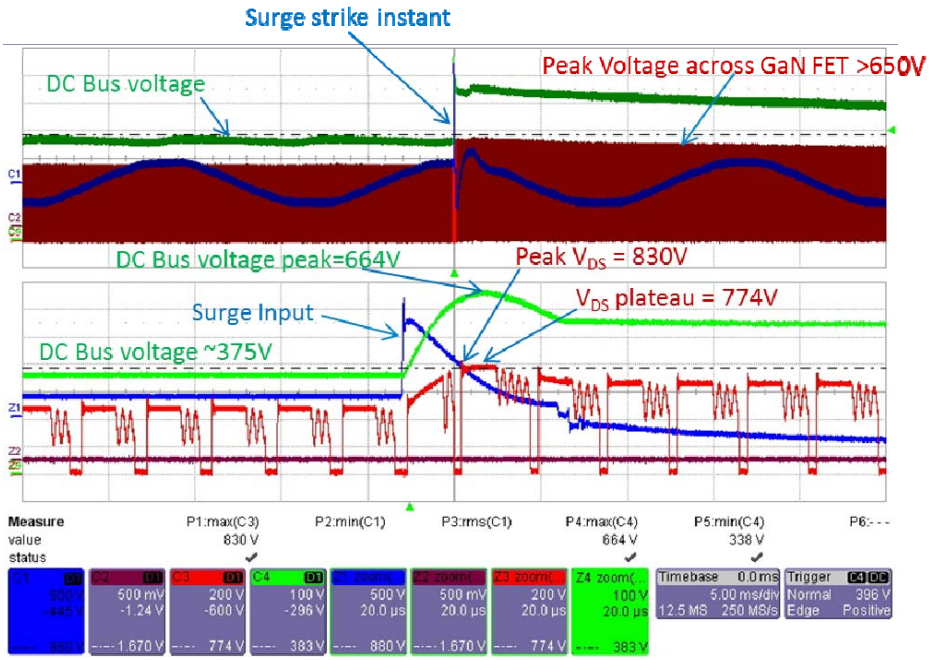

Fig. 7: Typical waveforms during a surge event highlighting the voltage across GaN FET

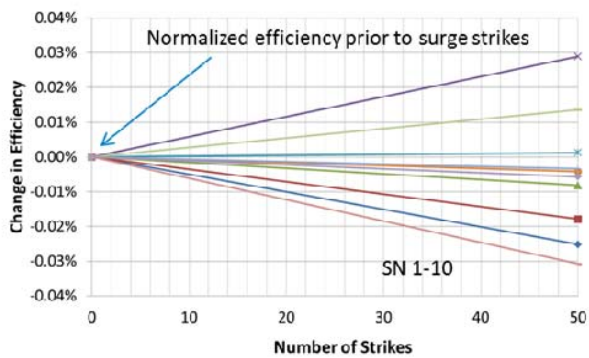

Fig. 8: Change in the flyback converter efficiency on 10 units before and after surge strikes with $90V_{AC}$ input.

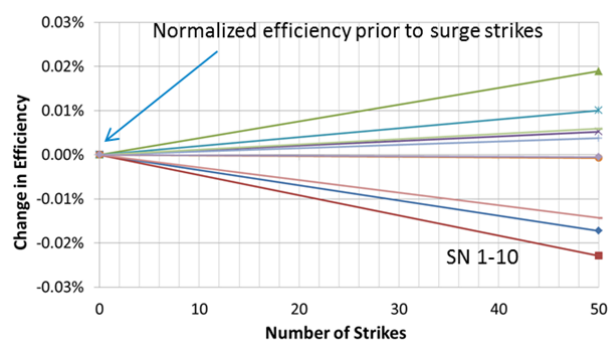

Fig. 9: Change in the flyback converter efficiency on 10 units before and after surge strikes with $264V_{AC}$ input.

surge strikes is shown in Figs. 8 and 9 corresponding to $90V_{AC}$ and $264V_{AC}$ inputs. Negligible change in efficiency is observed under both line conditions, and the minor variation observed from part to part is within the measurement accuracy of the set-up.

The above results clearly validate the ability of GaN FETs to withstand surge events despite not being avalanche rated like Silicon MOSFETs. Our experiment subjected the GaN FET to 830V peak voltage, providing sufficient margin above the datasheet specification of 750V maximum (non-repetitive) ensuring robust operation in the field.

IV. Performance Under Line Swell Conditions

Another factor to be considered while designing a power supply is its exposure to brief periods of line swell conditions affecting the distribution system caused by events such as shutting off loads, switching capacitor banks on or due to system fault conditions. Line voltage swell can result in high voltage across the GaN FET for a longer duration than in the case of the single surge strike considered previously.

In this work, we have evaluated the performance of the GaN FET under momentaneous swell condition based on the IEEE standard 1159-2009 [9] which defines the typical duration of a swell from 3s to 1min with an amplitude of 1.1 to 1.4 times the nominal value.

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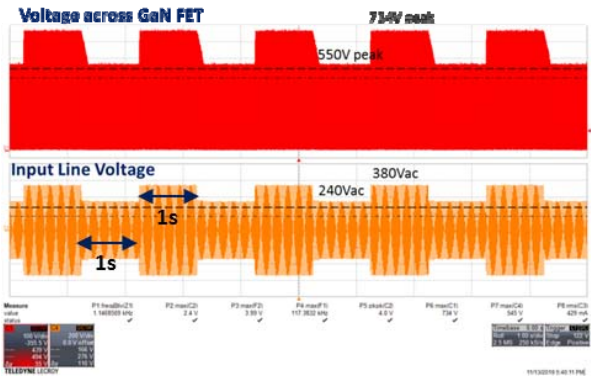

Fig. 9: Snapshot of the varying input voltage and voltage across the GaN FET for a period of 10s (out of 1Hr tested) under the line swell test.

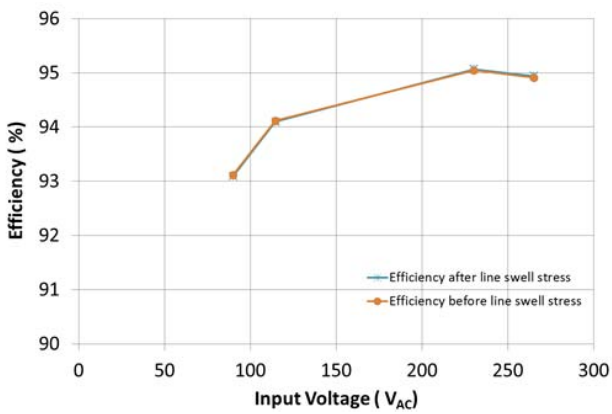

Fig. 10: Flyback converter efficiency across input voltage before and after the line swell test.

In our experiment, we subjected the part to even higher peak amplitude of 1.6 times the nominal value in order to ensure sufficient margin in real life operation in the field, and the input voltage applied to the converter was switched from the nominal condition of $240V_{AC}$ to $380V_{AC}$ every other second for a total duration of one hour. A snapshot of the varying input under this test for 10s along with the resultant voltage across the GaN FET is shown in Fig. 9. Peak voltage across the GaN FET reached 734V, well above the nominal operating voltage of 650V whenever input line swell reached $380V_{AC}$.

Fig. 10 shows a chart comparing the converter efficiency between pre and post line swell stress over $90V_{AC}$ - $264V_{AC}$ input condition. No degradation in converter efficiency was observed as a result of the application of the specified line swell stress test for one hour, once again confirming the robustness of GaN FETs enabling them to withstand high voltage conditions encountered in the field.

V. Conclusion

In this paper, we have evaluated the surge-robustness of a flyback power supply based on GaN FET for the first time, and demonstrated stable converter performance under standard surge conditions. We also demonstrated its robust performance under extended line swell conditions.

Our result dispels a major concern regarding the performance of GaN FETs under surge and other abnormal high input voltage conditions due to their lack of avalanche rating, thereby paving the way for wide-spread adoption of GaN in various cost-sensitive, high-volume consumer electronics applications.

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